
Algorithm 1 DAFB-CLS Forward Pass

Require: Image $\mathbf{x}$, frozen ViT Φ , layers $\mathcal{L} = \{3, 6, 9, 12\}$

Ensure: Decoupled representation $\mathbf{C}$

- 1: $\{\mathbf{x}^\ell\}_{\ell \in \mathcal{L}} \leftarrow \Phi.\text{extract}(\mathbf{x})$
 - 2: $\mathbf{c}_{\text{cls}} \leftarrow \mathbf{x}_0^L$
 - **1. Multi-cue foregroundness scoring**
 - 3: $S_i \leftarrow \text{FreqStability}(\mathbf{x}^\ell)$
 - 4: $D_i \leftarrow \text{DepthConsistency}(\{\mathbf{x}^\ell\})$
 - 5: $A_i \leftarrow \text{SemanticAlign}(\{\mathbf{x}^\ell\}, \mathbf{c}_{\text{cls}})$
 - 6: $C_i \leftarrow \text{SpatialCompact}(\cdot)$
 - 7: $\mathbf{F} \leftarrow \text{MLP}(\sum_k w_k [S_i, D_i, A_i, C_i])$
 - **2. Adaptive soft masking**
 - 8: $\tau \leftarrow \text{MLP}_\tau(\mathbf{c}_{\text{cls}})$
 - 9: $m_i^F \leftarrow \sigma((F_i - \tau)/T)$
 - 10: $m_i^B \leftarrow (1 - m_i^F) \cdot (1 - u_i)$
 - **3. Dual CLS aggregation with depth attention**
 - 11: **for** $\ell \in \mathcal{L}$ **do**
 - 12: $\mathbf{B}_\ell^F \leftarrow \sum_i m_i^F \mathbf{x}_i^\ell / \sum_i m_i^F$
 - 13: $\mathbf{B}_\ell^B \leftarrow \sum_i m_i^B \mathbf{x}_i^\ell / \sum_i m_i^B$
 - 14: **end for**
 - 15: $\beta^F \leftarrow \text{softmax}(\mathbf{q}_F^\top \cdot \text{RMSNorm}(\{\mathbf{B}_\ell^F\}))$
 - 16: $\beta^B \leftarrow \text{softmax}(\mathbf{q}_B^\top \cdot \text{RMSNorm}(\{\mathbf{B}_\ell^B\}))$
 - 17: $\mathbf{C}_F \leftarrow \sum_\ell \beta_\ell^F \mathbf{B}_\ell^F$
 - 18: $\mathbf{C}_B \leftarrow \sum_\ell \beta_\ell^B \mathbf{B}_\ell^B$
 - **4. Task-adaptive fusion**
 - 19: $g \leftarrow \sigma(\text{MLP}_g([\mathbf{C}_F, \mathbf{C}_B, \mathbf{c}_{\text{cls}}]))$
 - 20: $\mathbf{C} \leftarrow g \cdot \mathbf{C}_F + (1 - g) \cdot \mathbf{C}_B$
 - 21: **return** $\mathbf{C}$
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